Multi-Modal Agentic AI System for Medical Diagnosis and Triage

Project Pipeline Workflow - Victor Oluwadare - Code Review Manager

Team GitHub Repository: https://github.com/abhishek7112000/Week-4-Team-9

Project Overview

This project constructs an autonomous, multi-modal AI agent that diagnoses chest and cardiovascular conditions by sequentially analysing patient-submitted symptoms, chest X-ray images, and structured clinical records. Three machine learning models are built, evaluated, and compared before being integrated into a LangGraph.js agent deployed as a Next.js web application.

Datasets and model assignments:

Model	Type	Dataset	Output
Model 1 — ChestVision	Transfer Learning (fine-tuned via timm)	NIH ChestX-ray14	14-label X-ray classification — ViT-B/16 selected (ONNX)
Model 2 — ClinicalFusion	Hybrid (MLP + frozen BioClinical-ModernBERT-large)	Synthea Synthetic EHR	Diagnosis category + severity score (ONNX)
Model 3 — OpenMed QA	Pre-trained + transfer learning	Medical QA Dataset (HF)	Professional patient-facing response (HF Inference)
Agent LLM — MedGemma	Instruction-tuned (no fine-tuning)	N/A — used as-is	Clinical synthesis and DiagnosisReport JSON (HF Inference)
RAG — Encyclopedia	Retrieval-augmented generation	A-Z Medical Encyclopedia	Double-check layer; ChromaDB knowledge base

1. Data Pre-Processing and Feature Engineering

Dataset 1 — NIH ChestX-ray14 (Model 1)

112,120 frontal chest X-ray images labelled with 14 thoracic conditions. The official NIH train/test patient-level split is used to prevent data leakage.

- Resize all images to 224×224 (ResNet50V2, ViT-B/16, EfficientNetV2B0 — unified size); replicate greyscale channel to 3-channel RGB; apply ImageNet normalisation (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
- Training augmentation: RandomResizedCrop(224, scale=0.85–1.0, ratio=0.95–1.05), RandomHorizontalFlip(p=0.5), RandomRotation(±10°), ColorJitter(brightness=0.2, contrast=0.2), RandomAffine(translate=0.05). No vertical flip (anatomically invalid).
- Class imbalance: BCEWithLogitsLoss with per-class pos_weight = (N^- / N^+), capped at 20 to prevent loss explosion on rare classes. Single correction strategy — WeightedRandomSampler deliberately not combined with pos_weight to avoid double-correction.

Dataset 2 — Synthea Synthetic EHR (Model 2)

Synthetic patient records filtered to cardiovascular and respiratory ICD-10 codes (I21, I25, I50, I48, J18, J44, J45). Feature engineering produces two parallel inputs for the hybrid model.

- Tabular branch input: standardised demographics, pivoted lab values (troponin, BNP, creatinine, Hgb, CRP), vital signs (latest and 90-day mean), binary medication flags, comorbidity flags, encounter counts, and derived terms (pulse pressure, shock index).
- Textual branch input: template-generated clinical summary per patient (e.g. "58-year-old male, prior MI, elevated troponin, on beta-blocker") tokenised for the frozen BioClinical-ModernBERT-large encoder (max_length=512). Missing values: group-median imputation with binary indicator columns.
- Labels: 5-class diagnosis category (Cardiac / Respiratory / Overlap / Hypertensive Crisis / Other) and ordinal severity score 0–3 derived from troponin elevation, SpO2 < 94%, SBP > 180, and hospitalisation history.

Dataset 3 — Medical QA Dataset (Model 3)

- Load Malikeh1375/medical-question-answering-datasets from HuggingFace Hub. Tokenise Q→A pairs with the OpenMed tokeniser (max_length=512). Retain pairs relevant to chest and cardiovascular conditions for domain alignment.

RAG Knowledge Base — A-Z Medical Encyclopedia

- Chunk encyclopedia into 512-token segments (64-token overlap) using LangChain RecursiveCharacterTextSplitter. Embed each chunk with the OpenMed encoder CLS token (768-dim) and store in ChromaDB (HNSW index). Used by the agent's RAG node at runtime only.

2. Model Training and Improvement

Model 1 — ChestVision (Transfer Learning — Fine-Tuned via timm)

Three pretrained architectures are fine-tuned using ImageNet weights via timm and compared: ResNet50V2, ViT-B/16, and EfficientNetV2B0. ViT-B/16 is selected as the final deployed model based on highest validation AUC. (Note: ViT-B/16 from scratch was discarded — ~100k images is insufficient for scratch Vision Transformer training.)

	ResNet50V2 (timm: resnetv2_50)	**ViT-B/16 ★ Selected (timm: vit_base_patch16_224)**
Architecture	ResNet50V2 pretrained ImageNet weights via timm; fresh head: GlobalAvgPool → Dropout(0.5) → Linear(num_features, 14) → Sigmoid. ~25M params.	ViT-B/16 pretrained ImageNet weights via timm; 12 transformer blocks, 16×16 patches; fresh head: Linear(768, 14) → Sigmoid. ~86M params.
Loss	BCEWithLogitsLoss with pos_weight (N^-/N^+, capped at 20)	BCEWithLogitsLoss with pos_weight (N^-/N^+, capped at 20)
Optimiser / LR	AdamW; Phase 1 (head frozen): lr=1e-3 × 5 epochs → Phase 2 (full fine-tune): lr=1e-5 × 15 epochs	AdamW; Phase 1 (head frozen): lr=1e-3 × 5 epochs → Phase 2 (full fine-tune): lr=1e-5 × 15 epochs
Batch / Epochs	32 / 20 total (5+15); early stopping patience=6	16 / 20 total (5+15); early stopping patience=6
ONNX export	~98 MB (fits Vercel 250 MB without quantisation)	~330 MB raw; INT8 quantisation → ~85 MB (fits Vercel 250 MB)

EfficientNetV2B0 (timm: tf_efficientnetv2_b0) is also trained and compared as a third architecture. ~7M params; pretrained ImageNet weights; two-phase schedule identical to ResNet50V2 and ViT-B/16; INT8 quantised ONNX ~30 MB. Included in the cross-model comparison table; not selected as the final deployed model.

- Hyperparameter tuning: Optuna (50 trials, Bayesian), objective = macro mean AUC across 14 classes. Search space: lr [1e-4, 5e-3], dropout [0.3, 0.6], pos_weight_cap [10, 30], weight decay [1e-5, 1e-3]. MedianPruner halts unpromising trials at epoch 10.

Model 2 — ClinicalFusion (Hybrid Model)

Two branches process the same patient in parallel; their outputs are concatenated and passed through a fusion classification network.

- Tabular MLP branch: Linear(input, 256) → BatchNorm → ReLU → Dropout(0.3) → Linear(256,128) → ReLU → Dropout(0.3) → Linear(128,64). All weights trained from scratch.
- Frozen BioClinical-ModernBERT-large branch: patient clinical summary passed through BioClinical-ModernBERT-large encoder with all weights frozen (requires_grad=False). Outputs 1024-dim CLS token projected to 256-dim. Optional: unfreeze top 2 layers after epoch 15 at lr=1e-6.
- Fusion: concatenate [256-dim tabular || 256-dim projected text] → Linear(512,256) → BatchNorm → ReLU → Dropout(0.4) → Linear(256,128). Two output heads: diagnosis (Softmax, 5 classes) and severity (Softmax, 4 classes). Combined loss: 0.6×CE(diagnosis) + 0.4×CE(severity).
- Training: AdamW lr=3e-4, batch=128, epochs=60 (early stop on weighted F1). Optuna tunes hidden dims, dropout, loss weight split. Ablation compares MLP-only, text-only, and full fusion.

Model 3 — OpenMed QA (Pre-Trained + Transfer Learning)

- Base model: OpenMed generative or extractive QA variant (MaziyarPanahi / OpenMed on HuggingFace). Fine-tuned on the Medical QA dataset using HuggingFace Trainer. Lower 60% of encoder layers frozen to preserve general biomedical knowledge.
- Optimiser: AdamW lr=2e-5, warmup_ratio=0.06, batch=16 (gradient accumulation ×4), epochs=5. After fine-tuning, model is pushed to HuggingFace Hub and served via the Inference Endpoints API.

3. Model Evaluation and Metrics

Model	**Primary Metrics**	**Targets / Thresholds**

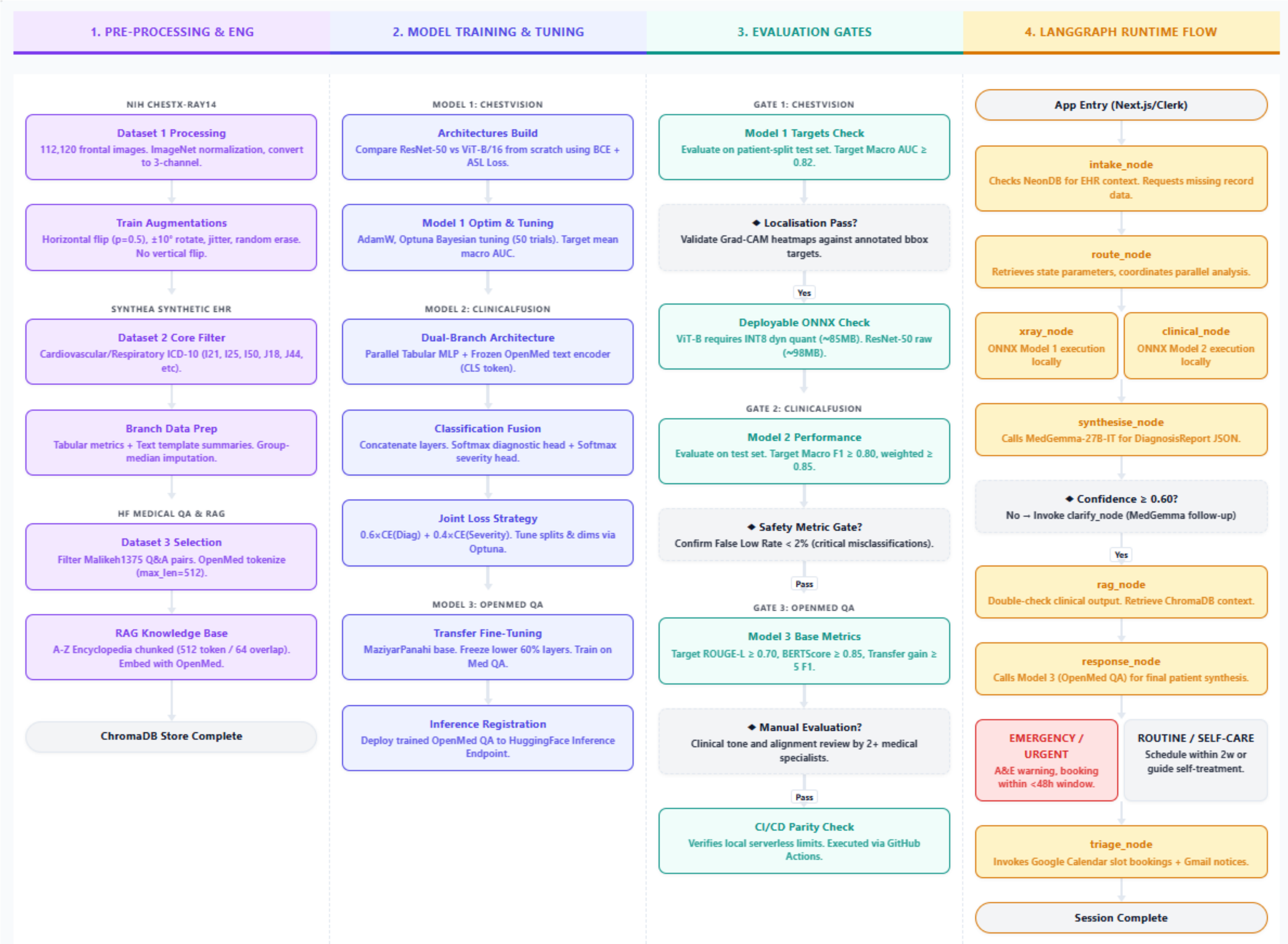
Model 1ChestVision	Mean AUC-ROC (14-class macro average)Per-class AUC-ROC, Average PrecisionF1 at optimal threshold (Youden's J)Grad-CAM localisation AUC (bounding boxes)ONNX CPU latency	Mean AUC ≥ 0.82 (CheXNet DenseNet-121 benchmark: 0.841)≥ 0.75 per class≥ 0.60 for top 5 conditionsReport vs annotated regions< 50 ms per image
Model 2ClinicalFusion	Macro F1 and weighted F1 (diagnosis)ROC-AUC one-vs-rest per classCohen's Kappa (severity)Severity MAEFalse low rate (safety metric)ONNX CPU latency	Macro F1 ≥ 0.80; weighted F1 ≥ 0.85≥ 0.90 per class≥ 0.75< 0.40< 2% (critical patients misclassified as low severity)< 20 ms per record
Model 3OpenMed QA	ROUGE-L, BERTScoreExact Match / F1 token overlapTransfer gain vs base modelClinical tone review (manual)	ROUGE-L ≥ 0.70; BERTScore ≥ 0.85F1 token ≥ 0.80Fine-tuned must exceed base by ≥ 5 F1 ptsPass team review (2+ members)

All three models are evaluated on held-out test sets with no overlap with training or validation data. A cross-model comparison table records primary metric, ONNX export size, and CPU inference latency for each variant. The ablation study for Model 2 (MLP-only vs text-only vs full hybrid) quantifies the contribution of each branch. Grad-CAM heatmaps for Model 1 and the confusion matrix for Model 2 are included in the final report to provide visual evidence of model behaviour.

4. Application Deployment

The application is built entirely in TypeScript. Python is used only for model training; no Python services run in production. ONNX models (Models 1 and 2) are bundled directly into Vercel serverless functions and executed via **onnxruntime-node**. Model 3 (OpenMed QA) and MedGemma-27B-IT are served via HuggingFace Inference Endpoints and called using **@huggingface/inference**.

Agent workflow: The LangGraph agent runs as a stateful directed graph. The patient enters symptoms and optionally uploads a chest X-ray via the AgentChat UI. The **intake_node** checks NeonDB for an existing clinical record (requesting upload if absent). The **route_node** invokes **xray_node** (Model 1 ONNX) and **clinical_node** (Model 2 ONNX) in parallel. Results are passed to the **synthesise_node**, which calls MedGemma-27B-IT to produce a structured DiagnosisReport. If confidence < 0.60, the **clarify_node** generates follow-up questions. The **rag_node** then queries ChromaDB to double-check findings against the medical encyclopedia. Finally, the **response_node** passes the report and RAG context to the fine-tuned OpenMed QA model to generate the patient-facing response. Triage level (EMERGENCY / URGENT / ROUTINE / SELF_CARE) determines whether **triage_node** books a Google Calendar appointment and sends Gmail confirmations.



[Diagram update required manually: (1) M1 box — replace "ResNet-50 vs ViT-B/16 from scratch + ASL" with "Fine-tune ResNet50V2 / ViT-B/16 / EfficientNetV2B0 via BCEWithLogitsLoss + pos_weight". (2) M2 box — replace "Frozen OpenMed encoder" with "Frozen BioClinical-ModernBERT-large". (3) ONNX box — add EfficientNetV2B0 INT8 ~30 MB.]

Figure 1: Pipeline Workflow